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MARKET

(Forms of Market, Producer's Equilibrium and Price and Output Determination in Perfect Competition)

QUICK REVISION

1. Market: Market is a system or a mechanism or an arrangement by which buyers and sellers of a commodity are brought in close contact for the purchase and sale of the commodity.

2. Forms of Market: There are four main forms of market:
(a) Perfect competition
(b) Monopoly
(c) Monopolistic competition and
(d) Oligopoly
(a) Perfect Competition: It means a form of market where there are large number of buyers and sellers of a commodity which is homogeneous in nature which is sold with no control over price by an individual firm at a fix price.
It is a market situation in which no individual firm can influence the market price. Market price is determined by the industry, i.e., by the collective action of all buyers and sellers.
Features:
(i) The number of sellers and buyers is very large. It is so large that no individual seller or buyer can influence the market price of the commodity.
Implications: Buyers and sellers of the product are just price takers. A competitive firm faces a perfectly elastic demand curve. Also have no bargaining power in the market.
(ii) All sellers (or firm) sell homogeneous products.
Implication: This feature ensures uniform price in the market.
(iii) The entry or exit of firm in the long run is free.
Implication: This feature ensures that there are neither abnormal profits nor losses by any firm in the long run.
(iv) Buyers and sellers have perfect knowledge about the market.
Implication: Uniform price will prevail in the market.
(v)Factors or production are perfectly mobile.
Implication: This feature ensures the uniform cost structure of all firms.
(vi) Transport costs are zero.
Implication: Transport costs do not affect the market price.

Pure Market: Pure competition exists when there are a great number of sellers whose products are all identical and no one seller controls a large enough part of the total output to exert an appreciable influence on price. Thus it may be concluded that pure competition is the state of market that satisfies following three conditions:
(i) Large number of buyers and sellers.
(ii) Homogeneous product.
(iii)Free entry and exit of firms.

(b) Monopoly: Monopoly is a market situation in which there is only one firm producing a commodity. The commodity has no substitutes in the market. Indian Railways is an example of monopoly.
Features:
(i)There is only one seller (or producer) of a product in the market.
Implication: Firm is a price-maker. It usually exploits the buyers by charging a high price of its product.
(ii) There are no close substitutes of the product in the market.
Implication: Buyers have to purchase the commodity from the monopolist or go without it.
(iii) There are barriers to the entry of new firms.
Implication: Firm earns abnormal profits in the long run.
(iv) Firm has full control over price.
Implication: Firm is price market.
(v) Price Discrimination: The act of selling the same product at different prices to different buyers is known as price discrimination. Monopolist can easily do price discrimination because he is the only seller of the product in the market. If a monopolist adopts the policy of price discrimination the situation is called discriminating monopoly.

(c) Monopolistic Competition: Monopolistic competition is a form of market which like perfect competition has large number of sellers and allows the firm the freedom of entry and exit and unlike perfect competition it has product differentiation and partial control over the price of the product.
Features:
(i) There is a large number of sellers and buyers.
Implication: No individual seller can influence the market price in a big way.
(ii) Product sold by different sellers are differentiated on account of different brand names, packing, colour, shape, customer service.
Implication: This feature makes monopolistic competition different from perfect competition. Firms are price makers.
(iii) Firms are free to enter or leave the industry.
Implication: All firms earn only normal profits in the long run.
(iv) Buyers and sellers do not have complete knowledge about the market.
Implication: Products sell at different price in the market.
(v) Firms spend a lot of money to promote their sales.
Implication: Market demand of the product rises. Selling costs also tends to raise the costs.

(d) Oligopoly:
Meaning: Oligopoly is a market structure characterized by a small number of large firms which are mutually dependent on each other for taking price and output decisions.
Classification:
(i) **Perfect and Imperfect oligopoly**: If the firms produce homogeneous products, it is called perfect

oligopoly. Steel industry is an example of perfect oligopoly. If the firms produce differentiated product, it is called imperfect oligopoly or differentiated oligopoly. Automobile industry is one example of imperfect oligopoly.

(ii) **Non-collusive and Collusive Oligopoly**: If the firms compete with each other, it is called non-collusive oligopoly. If the firms co-operate with each other rather than compete in setting the price and output, it is called collusive oligopoly.

Features:

(i) Oligopoly is composed of few large firms.

Implication: Action of each firm not only influences itself but also other firms are affected by it.

(ii) There is high degree interdependence among the firms.

Implication: This feature makes oligopoly different from other forms of market. Demand curve of a firm here is indeterminate.

(iii) There are barriers to the entry of firms.

Implication: Due to entry barriers, the number of firms in the industry remains small or limited.

(iv) There exists non-price competition. Firms normally are afraid of price-competition which may lead to price war. Product differentiations and advertisement are the two main forms of non-price competition.

Implication: Prices remain rigid in the market.

(v) Imperfect Knowledge: Buyers and sellers do not have perfect knowledge about the market of the product. Due to large amount of selling costs, consumers begin to know.

Producers' Equilibrium

1. **Who is a Producer?** Producer is a person or organization who takes decisions about to what to produce and how to produce to earn profit.

2. **Producer's Equilibrium**: A producer is said to be in equilibrium when he produces an output level at which his profits are maximum.

3. **Equilibrium Conditions**: There are two method for the determination of producer's equilibrium:

(i) TR-TC Approach: A producer will be in equilibrium at that level of output where the difference between TR-TC is maximum. If profit (TR-TC) is maximum at more than one output level, the output level beyond which profits starts declining in the equilibrium output.

(ii) MR-MC Approach: Usually in microeconomics MR-MC approach is used to determine the position of producer's equilibrium. The conditions of equilibrium according to this approach are:

(a) MC = MR (b) MC must cut MR from below.

Determination of Price and Output in a Perfect Competition

Firm is an individual enterprise which is a price taker & industry is a group of competing firms selling a product. Industry is a price maker.

1. Equilibrium in an Industry: In perfect competition no single firm or consumer can influence the price. Price is determined by the industry with the help of demand & supply forces. Demand curve of an industry is downward sloping representing aggregate demand of an economy, on the other hand is upward sloping aggregate supply of an economy. The determination of price will take at point where demand & supply curve are intersecting each other.

2. **Equilibrium of the Firm in Short-Run**: The general conditions for the equilibrium of the firms in the short-run will be:
(i) SMC = MR
(ii) SMC cuts MR from below.
But these conditions will not guarantee that the profit is earned by the firm. In order to know whether the firm is making profit or loss we need a short-run average cost curve. In this regard there are 3 conditions for firm's equilibrium. In short-run:
(i) AR > AC, Super Normal Profit
(ii) AR < AC, Loss and
(iii) AR = AC, Normal Profit. (Break-even point)
(iv) **Break-even point**: A firm breaks even when TR equals TC or AR equals AC. At this point, firm is earning zero economics profits. Firm earns only normal profit.
(v) **Shut-down point**: When price is just covering the AVC, it is called a shut-down point.
Firm will close down its business if price falls below the AVC. Thus, P = minimum AVC is the shut-down point in the short-run.
3. **Firm's Equilibrium in the Long Run**: Only Normal Profit
MC=MR = AR=LAC

1 MARK QUESTIONS

A. FILL IN THE BLANKS

PRODUCER'S EQUILIBRIUM AND PRICE AND OUTPUT DETERMINATION

1. TR= _____ x units of produced goods.

2. AR = _____ .

3. MR = _____ - _____ .

4. AR and MR curve coincide under _____ competition.

5. AR curve in monopoly is a _____ sloping curve.

6. If TR rises at a constant rate, MR is _____ .

7. When MR is negative, TR should _____ .

8. In perfect competition TR is rising at a _____ rate.

9. Producer is in equilibrium when the gap between _____ and _____ is maximum.

10. A producer is in equilibrium when MR is _____ to MC and next _____ is rising.

11. A firm earns normal profit when AR _____ AC.

12. At the point of equilibrium _____ must be rising.

13. TR-TC= _____ , TR = TC = _____ .

14. AC>AR > AVC= _____ , AR < SAC = AVC _____ .

15. AR SAC= _____ , AR = SAC = _____ .

16. Under imperfect competition when TR is maximum MR would be _____ .

17. A single unit to higher factors of production to produce goods is known as _____ .

18. ARMR = LMC = LAC= _____ .

19. Break-even point _____ .

20.[] _____ .

21. A producer is in equilibrium, where he gets _____ .

ANSWERS

1. Price	2. Price	3. TRA TRn-1	4. Perfect
5. Downward	6. Constant	7. Fall	8. Constant
9.TR, TC	10.Equal, MC	11.Equal to	12. MC
13. Profit, Normal Profit	14.No shut down, shut down	15.Super normal profit,Normal profit	16. Zero
17. Firm	18. Normal profit in the long run	19.AR = AC	20. TR-TC
21. Maximum Profit			

MARKET FORMS

1. _____ and _____ are forms of imperfect competition.

2. Due to large number of buyers and sellers, buyers and sellers have no control over _____.

3. Perfect competition deals with _____ products.

4. Due to _____perfect competition earns _____ profit in the long run.

5. There is no _____ cost in the perfect competition.

6. Under perfect competition _____ and _____ are same.

7. In short run, perfect competition can have normal profits, _____ or _____.

8. The entry to monopoly is _____.

9. Monopoly can arise due to _____ and _____.

10. AR and MR curves of monopoly slope _____.

11. A monopoly firm is a price _____ but perfect competition firm is a price.

12. A group of firms working together to secure a monopoly is called _____.

13. Product differentiation is an important characteristic of _____ competition.

14. Monopolistic competition incurs _____ selling cost.

15. Oligopoly firms have _____ price.

16. _____ and _____ are best examples of oligopoly in India.

17. On the basis of Co-operation, oligopoly can be _____ or _____.

18. On the basis of nature of product, oligopoly can be _____ or _____.

19. Both oligopoly and monopolistic competition try to practice _____ competition.

20. There is no difference between a firm and an industry in a _____.

ANSWERS

1. Monopolistic competition, oligopoly	2. Price	3. Homogeneous products	4. Free entry and exit, Normal profit
5. Selling cost	6. AR, MR	7. Super normal profit, super normal loss	8. Closed
9. Nature, Government	10. Downwards	11. Maker, Taker	12. Cartle
13. Monopolistic	14. High	15. Rigid	16. Petroleum market, Mobile phones market
17. Collusive, oligopoly	18. Homogeneous, Differentiated	19. Monopolistic	20. Monopoly market.

B. TRUE OR FALSE

1. Under monopoly a firm can sell all goods at single price.
Ans. **[False]** Because a monopoly firm can sell the units of a goods at different prices.

2. In perfect competition, price is determined independently by a firm.
Ans. **[False]** In perfect competition price is determined by industry and firm is only a price taker.

3. Under monopolistic competition there is only one seller.
Ans. **[False]** Because there are large number of buyers and sellers in monopolistic competition.

4. In perfect competition products sold are identical.
Ans. **[True]** As price in perfect competition is constant so the product has to be same in all respects.

5. In oligopoly market, there are large number of buyers and sellers.
Ans. **[False]** In oligopoly market there are a few big sellers and a large number of buyers.

6. In monopolistic market price discrimination can be easily done.
Ans. **[False]** Because price discrimination is only possible in a monopoly market where seller has full control over the price.

7. In monopoly, firm is itself an industry.
Ans. **[True]** As monopoly firm itself constitutes industry due to non-existence of any other firm dealing in the same product.

8. Equilibrium price under perfect competition is determined by each firm with its demand and supply of the product.
Ans. **[False]** Equilibrium price is determined by market forces of demand and supply in the industry.

9. Firm's demand curve under monopolistic competition is more elastic under monopoly.
Ans. **[True]** Because under monopolistic competition close substitutes are available.

10. Firm's demand curve is indeterminate under oligopoly.
Ans. **[True]** Because there is high degree of interdependence between the firms and therefore a firm cannot be certain about nature and position of its demand curve.

11. There is no selling cost in perfect competition and monopoly market.
Ans. **[True]** Because under perfect competition, homogeneous products are sold at a uniform price and in monopoly, product has no close substitutes in the market.

12. At the state of producer's equilibrium MC curve of the firm should be rising.
Ans. **[True]** If MC is falling, then it is possible to increase profits by producing more. So MC should be rising at the state of producer's equilibrium.

13. To maximize the profits of a firm, the only condition needed is equality between marginal cost and marginal revenue.
Ans. **[False]** Profits are maximized when two conditions are satisfied: (i) MC = MR (ii) MC must cut MR from below.

14. If MC = MR at two output levels, then any one of the output level can be taken as a state of producer's equilibrium.
Ans. **[False]** Only that output level is the state of producer's equilibrium when MC becomes greater than MR after the equilibrium.

15. Excess of MR over MC is always better than equality between the two in order to achieve the equilibrium for a producer.
Ans. **[False]** Producer will not be at equilibrium when MR is more than MC as it will be possible to increase profits by producing more. So equality between MR and MC is a better situation.

C. MULTIPLE CHOICE QUESTIONS

PRODUCER'S EQUILIBRIUM

1. A producer is in equilibrium, where he gets:
(a) maximum satisfaction (b) maximum profits
(c) minimum costs (d) any of these

2. For firm's equilibrium with total revenue concept, it is essential that:
(a) TR=TC (b) TC cuts TR from below
(c) maximum difference between TR and TC (d) none of these

3. According to Marginal concept, a producer is in equilibrium where:
(a) MC - MR (b) MC cuts MR from below

(c) both (a) and (b)
(d) none of these

4. Producer's equilibrium indicates about:
(a) ideal level of output
(b) ideal price of the product
(c) expected profit
(d) all of these

5. A competitive firm is in equilibrium when it gets:
(a) super normal profits
(b) normal profits
(c) expected profit
(d) all of these

6. A competitive firm will be in equilibrium when:
(a) MR = MC and MC cuts MR from below
(b) AR = AC, AC cuts AR from above
(c) TR=TC
(d) none of these

7. A competitive firm is in equilibrium during long period:
(a) when MC = MR = AR = AC
(b) when AC > AR
(c) when AR< AC
(d) none of these

8. Which of the following statement is false about producer's equilibrium?
(a) When the difference between TR and TC is minimum
(b) MR = MC
(c) Producer gets maximum profits
(d) Producer does not want any changes till circumstances are unchanged.

ANSWERS

1. (b)	2. (c)	3. (c)	4. (d)	5. (d)	6. (a)	7. (a)	8. (a)

FORMS OF MARKET

1. Which of the following is a feature of market?
(a) Buyers and sellers of region
(b) A price of the product
(c) Any means of communication between buyers and sellers
(d) All of these

2. Which type of market it is, which has a large number of buyers and sellers, selling homogeneous goods at a single price?
(a) Imperfect market
(b) Perfect market
(c) Monopoly
(d) Monopolistic competition

3. Which of the following is a market situation that has single seller, having complete control over supply?
(a) Perfect competition (b) Monopoly
(c) Oligopoly (d) Monopolistic competition

4. In which type of market, a firm is a price maker?
(a) Perfect competition (b) Imperfect competition
(c) Monopoly (d) None of these

5. In which type of market, a firm is price taker?
(a) Competitive market (b) Monopoly
(c) Oligopoly (d) None of these

6. In which type of market AR and MR are downward sloping and more elastic?
(a) Competitive (b) Monopoly
(c) Monopolistic competition (d) Any of these

7. Which of the following market a firm has perfectly elastic demand curve (AR and MR)?
(a) Perfect competition (b) Monopoly
(c) Oligopoly (d) None of these

8. Which of the following market has product differentiation?
(a) Competitive market (b) Monopoly
(c) Monopolistic competition (d) Oligopoly

9. Which of the following market has price discrimination?
(a) Monopoly (b) Perfect competition
(c) Monopolistic competition (d) None of these

10. What type of market it is, which has a large number of buyers and sellers, selling close substitutes of the product
(a) Pure competitive firm (b) Monopoly
(c) Monopolistic competition (d) All of these

11. Which of the following costs help in promoting sales of the firms?
(a) Selling costs (b) Production costs
(c) Real costs (d) All of these

12. Which of the following market sells unique goods, having no substitute?
(a) Monopolistic competition (b) Monopoly
(c) Pure competition (d) None of these

13. Which of the following market actually exists in real life?
(a) Perfect competition (b) Monopoly

(c) Monopolistic competition (d) None of these

14. Which of the following markets, has a few sellers?
(a) Perfect competition (b) Monopoly
(c) Oligopoly (d) None of these

15. Which of the following market has less elastic demand (AR and MR)?
(a) Competitive market (b) Monopoly
(c) Oligopoly (d) None of these

16. Which of the following market has kinked demand curve?
(a) Monopoly (b) Monopolistic competition
(c) Oligopoly (d) None of these

17. In a competitive market, who takes pricing decisions?
(a) A firm (b) An industry
(c) Both (a) and (b) (d) None of these

18. Which of the following is not true about a market?
(a) Area of market can be a nation of the whole world
(b) Close contact between buyers and sellers
(c) Several goods are bought and sold
(d) Several buyers and sellers

19. Which of the following market has horizontal demand curve?
(a) Monopoly (b) Monopolistic competition
(c) Oligopoly (d) Perfect competition

20. How many firms are there in oligopoly market?
(a) Large number (b) A few
(c) Single (d) A many

21. Perfect oligopoly has:
(a) perfect knowledge about market (b) heterogeneous goods
(c) perfect mobility of goods (d) homogeneous goods

22. Oligopoly market has:
(a) independent character. (b) inter-dependence
(c) run by government (d) monopolistic nature

ANSWERS

1. (d)	2. (b)	3. (b)	4. (c)	5. (a)	6. (c)	7. (a)	8. (c)	9. (a)	10. (c)
11. (a)	12. (b)	13. (c)	14. (c)	15. (b)	16. (c)	17. (b)	18. (c)	19. (d)	20. (b)
21. (d)	22. (b).								

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : A perfect competitive firm earns normal profit in the long run.

Reason (R) : There is freedom of entry and exit of the firm in a perfect competitive market.

2. Assertion (A) : In monopolistic competition demand curve for a firm is highly elastic.

Reason (R) : In monopolistic market large number of sellers sell differentiated products which are the close substitutes of each other.

3. Assertion (A) : Selling cost plays very important role in a monopolistic market.

Reason (R) : In a Monopolistic market sellers sell differentiated goods which are perfect substitutes of each other.

4. Assertion (A) : In a monopoly market, firm and industry are the same.

Reason (R) : A monopoly market consists of only one seller or firm because there is only firm in the market, the firm is regarded as the industry

5. Assertion (A) : Under perfect competition, price is determined by equilibrium of demand and supply.

Reason (R) : In a perfect competition, market firm is a price taker and industry makes the price with the help of market forces, demand and supply.

6. Assertion (A) : Perfect competition prevails when the demand for the output of each producer is perfectly elastic.

Reason (R) : A single uniform price prevails under perfect competition which is determined by the interaction of demand and supply.

7. Assertion (A) : The state of the producer's equilibrium either reflects maximum profits or minimum losses.

Reason (R) : When MC > MR after equilibrium, it means producing more will lead to rise in profits.

8. Assertion (A) : A firm is at equilibrium when MR = MC and beyond that level of output, MC must be falling.

Reason (R) : MC curve should cut MR curve from below to achieve producer's equilibrium.

9. Assertion (A) : Product differentiation is the main feature of monopolistic competition.

Reason (R) : Under monopolistic competition, all the products are perfect substitutes to each other.

ANSWERS

1. (d)	2. (e)	3. (d)	4. (d)	5. (d)	6. (d)	7. (a)	8. (b)	9. (c)

2 MARK QUESTIONS

1. Define the term 'Market'.
Or What do you mean by 'market' in economics?
Ans. In economics, the term 'market' refers to an institutional arrangement through which the sellers and purchasers of any commodity or a service can come into contact with each other, and complete the act of sale and purchase.

2. Define a perfectly competitive market.
Ans. When there are a large number of sellers who sell similar products at a given price in the market, and where any individual seller has no control over the market price, we call it a perfectly competitive market.

3. What is mean by pure competition?
Ans. If in a market structure: (i) there are many buyers and sellers, (ii) all the firms produce and sell similar products and (iii) there remains no barrier either on entry into or exit from the market, we call it pure competition.

4. What is monopolistic competition?
Ans. When a large number of firms, each having a partial control over the market price, produce differentiated products and compete with each other to raise the market share, it is called monopolistic competition.

5. What is meant by oligopoly?
Ans. When a small number of firms control the total market supply of a particular commodity and when each of them possesses some market power, it is called an oligopoly market.

6. What is pure monopoly?
Ans. Pure monopoly refers to a market structure where only one firm constitutes the whole industry, and there is no substitute of the product produced by that firm.

7. What do you mean by product differentiation?
Ans. When different firms, particularly under monopolistic competition, sell substitute products under different brand names (say, different varieties of toilet soap), then it is called product differentiation.

8. In which market form are average revenue and marginal revenue of a firm always equal?
Ans. In a perfectly competitive market, the average revenue (AR) and the marginal revenue (MR) of a firm become equal since it sells at a given price.

9. Under which market form, a firm is a price-taker?
Ans. In a perfectly competitive market, an individual firm is a price-taker because it has to sell at a given price determined through market forces.

10. Under which market form, a firm is a price-maker?
Ans. In monopoly market, a firm is called a price-maker since it can influence market price through its control over the aggregate market supply.

11. In which market form, the products are homogeneous?
Ans. In a perfectly competitive market, the products are homogeneous in nature.

12. In which form of market there is product differentiation?
Ans. Product differentiation is observed under monopolistic competition.

13. In which form of market, a firm can follow price discrimination?
Ans. In a monopoly market, a firm can follow price discrimination, i.e., it can sell the same product at different prices to different groups of buyers.

14. Under which market, a firm faces the market demand curve?
Ans. In monopoly market, a firm that represents the whole industry, faces the market demand curve.

15. Mention any source of monopoly power of a firm.
Ans. The Patent Right or a Trade license is considered to be a source of monopoly power of a firm.

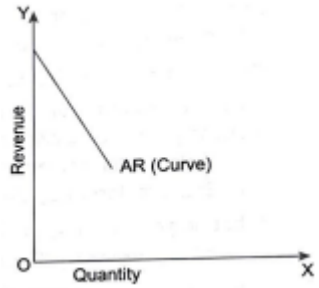
16. Under which form of market, the firms are engaged in non-price competition?
Ans. The firm are engaged in non-price competition particularly under monopolistic competition.

17. What do you mean by non-price competition?
Ans. When different firms try to increase their respective market shares, not by reducing the product price but by changing the brand names or spending more on advertisement, it is called non-price competition.

18. Under which market form, there remains a strategic intern-dependence between the production decisions taken by the firms?
Ans. In an oligopoly market, there remains a strategic inter-dependence between the production decisions taken by the firms.

19. What do you mean by collusive oligopoly?
Ans. When the firms in an oligopoly market enter into a collusive agreement for the fixation of price, we call it collusive oligopoly.

20. What is the shape of the demand curve faced by any monopoly firm?
Ans. Since the monopoly firm is the sole seller of a product, it faces the Y market demand curve. This demand curve is negatively sloped and indicates relatively inelastic demand for the product (because of the absence of close substitutes).



21. What do you mean by natural monopoly?
Ans. Sometimes the limited size of a market may allow the existence of only one firm of optimal size. The firm will then be capable of producing the product in a large-scale at the lowest average cost. The scale of production becomes so large that it can satisfy the whole market. This type of monopoly power is called the natural monopoly.

22. Why are bathing soaps sold in different shapes, colours, fragrances and sizes?
Ans. Since there is monopolistic competition in the bathing soap market, so different firms in this market want to grab the market share through product differentiation. Hence, different brands of bathing soaps are sold in different shapes, colours, fragrances and sizes. It creates an impression upon the minds of the consumers that these products are different. Though there may not be any real difference between them.

23. What is the basic difference between the demand schedules faced by a firm under monopolistic competition and under monopoly market?
Ans. In the monopoly market, the seller faces the market demand schedule that shows relatively inelastic demand. This happens due to the non-existence of close substitutes in the monopoly market.
However, the perceived demand schedule faced by each firm under monopolistic competition shows relatively elastic demand for the product. This happens due to the existence of many substitute products.

24. Indicate any two similarities between monopolistic competition and perfect competition.
Ans. Two important similarities between monopolistic competition and perfect competition are mentioned below:
(a) Large number of sellers: In both these forms of market structures, there exist a large number of sellers.
(b) Free entry and exit: In both these markets, there remains no barrier either upon the entry of a firm into the market or upon its exit from the market.

25. State any two factors affecting the type of market for a commodity.
Ans. The two factors given ahead affect the type of market for a commodity:
(i) Number of sellers in the market: If there remains a single seller of the commodity, it is called a monopoly market.
(ii) Extent of control over the market price: If any single seller does not have any control over the market price, we call it a perfectly competitive market.

26. Give two reasons why monopolistic competition is the most realistic form of a market.
Ans. In reality, pure monopoly and perfectly competitive market structures are rarely observed.
(i) Since, monopolistic competition is a blending of monopoly and competitive market structures, this seems to be the most realistic form of market.
(ii) We see such a market form in various consumer goods market. Here, each firm has some market power and they compete with each other to grab the market share through product differentiation (particularly through advertisements). For instance, monopolistic competition in the toothpaste and toilet soap market of India.

27. In which form of market is the seller a price-maker? Justify your answer.
Ans. In a monopoly market, the seller is considered as a price-maker. This is because of the fact that:
(i) The seller being the sole supplier of the product, can influence the market price; and
(ii) There is no substitute product in the market since the entry of any new firm is restricted.

28. Which features of monopolistic competition are monopolist in nature?
Ans. (i) Control over price (ii) Downward sloping demand curve.

29. Give the similarities between monopoly and monopolistic competition.
Ans. Monopolistic competition is a blending of monopoly and competitive market conditions, thus some features of monopoly market are also present in monopolistic competition:
(i) Each firm has some monopoly power in this.
(ii) Each firm faces negatively sloped demand curve or AR curve.

30. What do you mean by producer's equilibrium?
Ans. By producer's equilibrium we mean a situation where the producer produces a profit-maximizing level of output.

31. What are the two approaches for analyzing the producer's equilibrium?
Ans. These two approaches are: (i) Total revenue-Total cost approach, and (ii) Marginal revenue-Marginal cost approach.

32. How can you determine the profit schedule (or the profit curve) of a firm?
Ans. The profit schedule (or the profit curve) of a firm can be determined from the vertical distance between the total revenue (TR) and the total cost (TC) schedules.

33. What is gross profit?
Ans. Gross profit = Total revenue - Total variable cost of a firm.

34. What is net profit?
Ans. Net profit = Gross profit - Total fixed cost = (Total revenue - Total variable cost) - Total fixed cost of a firm.

35. What is the relation between MC and MR at producer's equilibrium?
Ans. At producer's equilibrium, the marginal cost (MC) and the marginal revenue (MR) become equal, where MC shows a rising trend.

36. If MC > MR, at any particular level of output, what will be its impact on producer's behaviour for maximizing its profit?
Ans. The producer will reduce the production to maximize its profit.

37. What is the first condition for producer's equilibrium?
Ans. The first condition (or the necessary condition) for producer's equilibrium is the equality between marginal marginal revenue (ie., MCMR) at the profit-maximizing level of output.

38. What is the second condition of producers equilibrium under perfect competition?
Ans. The second condition for producer's equilibrium under perfect competition is that the Marginal Cost (MC) must be rising at the profit-maximizing level of output.

39.What is the third condition for producer's equilibrium?
Ans. The third condition for producer's equilibrium is that the product price (P) must be greater than or at least equal to the average variable cost (AVC) of production.

40. What do you mean by break-even point?
Ans. When the producer's equilibrium shows the equality between average product price (P) and average cost (AC) of production, then it is called a break-even point.

41. What do you mean by a shut-down point?
Ans. If at producer's equilibrium, average product price (P) is just equal to the average variable cost (AVC), then it is called as the shut-down point.

42. When does a producer incur short-run loss?
Ans. A producer incurs short-run loss if the equilibrium price (P) is less than the average cost (AC) of production.

43. What do you mean by a super normal profit?
Ans. If the total revenue (TR) of a producer becomes greater than its total cost (TC) of production, which includes normal profit, then the excess profit is considered as the super-normal profit of the firm.

44. What type of profit is earned by a firm in the long-run under perfect competition?
Ans. In the long-run, a firm under perfect competition earns only normal profit.

3-4 MARKS QUESTIONS

1.(i) Define a monopoly market.
(ii) Give two examples of this market.
Ans. (i) If there is a single firm which produces and sells a product having no close substitute, this form of market structure is referred to as the monopoly market.
(ii) The Calcutta Electric Supply Corporation (CESC) enjoys the monopoly power in supplying electricity power at Kolkata, and the Indian Railways department enjoys a monopoly power in providing railways services in India.

2. Mention any four features of perfectly competitive market.
Ans. We can mention the following four important features of the perfectly competitive market:
(i) There are a large number of sellers and buyers in this market;
(ii) All the firms (the sellers) produce and sell a homogeneous commodity;
(iii) All the firms sell their products at a given price determined by the market forces of supply and demand; and
(iv) There is no barrier either upon the entry of new firms or on the exit of existing firms.

3. Mention any three features of the monopoly market.
or State three important characteristics of monopoly market.
Ans. Three important features (or characteristics) of the monopoly market are mentioned below:
(i) There remains a single seller and many buyers of a product in this market.
(ii) The product produced and sold by the monopoly firm has no close substitute.
(iii) Entry of new firms is restricted in this market.

4. Mention four features of monopolistically competitive market.
Ans. Four important features of a monopolistically competitive market are noted below:
(i) There are a large number of sellers and buyers in this market. substitutes but they are differentiated (say,

through a change in packaging, designing, brand names, etc.)
(ii) There remains no restriction either on the entry of a new firm into the market or upon the exit of a firm from the market.
(iv) Each firm spends a huge amount on advertisement with the objective of capturing the market share from the rival firms.

5. Indentify the market forms for the seller of 'A' and 'B' given the following information. Give reasons for your answers:

Output sold (unit)	Price of A(₹)	Price of B (₹)
10	5	5
20	5	4
30	5	3

Ans. In case of commodity 'A', there remains perfect competition since the seller of this commodity sells it at a given price(₹5).
However, in case of commodity 'B' there remains imperfectly competitive market (say, monopolistic competition) because the seller can sell more only at a lower price. Here, the seller faces the demand schedule which shows inverse relation between the price and quantity demanded.

6. In which form of market is the seller a price taker? Justify your answer.
Ans. In a perfectly competitive market structure the seller is price-taker because:
(i) The competitive price is determined by the market forces of supply and demand, and
(ii) Each seller has no individual control over that fixed market price. The seller has to accept that given price.

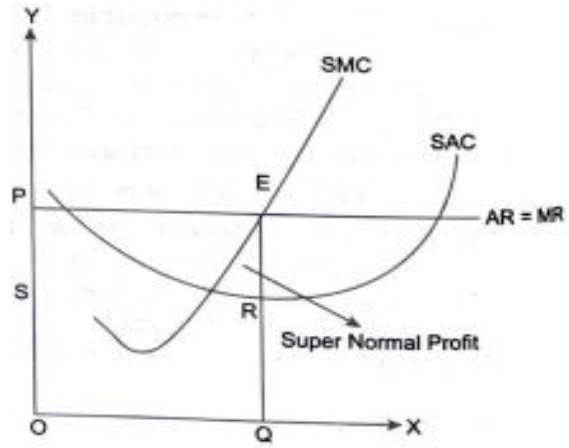
7. Mention any three features of the oligopoly market.
Ans. The following are the three main features of an oligopoly market:
(i) In this market, a small number of rival firms dominate the industry;
(ii) The production and marketing strategies of the rival firms are interdependent in nature and it is very difficult for a firm to estimate the probable counter strategy to be followed by a rival firm in response to its own strategy and
(iii) Here, the firms often enter into a collusive agreement with regard to the fixation of price through the formation of a cartel.

8. Give the similarities between perfect competition and monopolistic competition.
Ans. Similarities between perfect competition and monopolistic competition:
(i) In both the markets no. of buyers and sellers are large.
(ii) There are no restrictions on the entry into or exit of firms from the market.
(iii) In both the markets the principal goal of the firm is to maximize the profit.
(iv) Each firm earns only normal profit in the long-run.

9. Explain how short-run equilibrium is attained by a perfectly competitive firm earning super normal profit.
Ans. The general conditions for the equilibrium of the firms in the short-run will be:
(i) SMC = MR
(ii) SMC cuts MR from below.
But these conditions will not guarantee that the profit is earned by the firm. In order to know whether the firm is making profit or loss we need a short-run average cost curve. In this regard there are 3 conditions for firm's equilibrium. In short-run:

(i) AR > AC, Super Normal Profit
(ii) AR<AC, Loss
(iii) AR = AC, Normal Profit.

Since, in the shot-run, entry of new firm in the industry is not possible. There it becomes possible for a firm to earn super normal profit in the situation of equilibrium, which is shown in the graph in which AR > AC and the difference between AR & AC shows the profit earned by a firm in the short-run. Thus, in case of super normal profit:



Short Run: Super Normal Profit AR > AC

AR = MR = SMC > SAC

In the given graph TR = OPEQ

TC= OSRQ

Profit=TR-TC= PESR

10. Explain with the help of a well labelled diagram how a perfectly competitive firms earn normal profit in the short- run.

Ans. The general conditions for the equilibrium of the firms in the short-run will be:
(i) SMC = MR
(ii) SMC cuts MR from below.

But these conditions will not guarantee that the profit is earned by the firm. In order to know whether the firm is making profit or loss we need a short-run average cost curve. In this regard there are three conditions for firm's equilibrium. In short-run:

AR > AC, Super Normal Profit

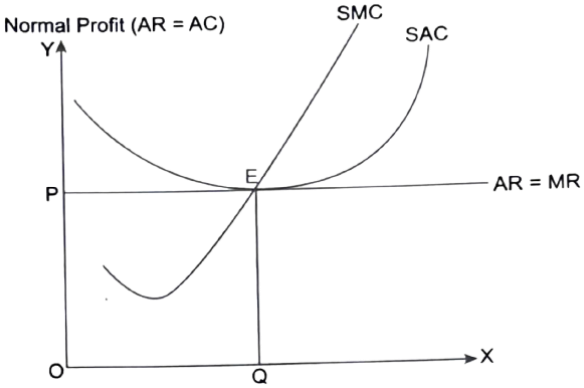
AR <AC, Loss

AR=AC, Normal Profit.

It is possible in the short-run for a perfect competition to earn normal profits. In this situation AR = AC and it is the condition of no profit, no loss which is also known as break-even point for a firm. Hence, in case of normal profit

AR=MR = SMC = SAC

In the given graph TR = OPEQ, TC = OPEQ and TR = TC. Thus, producer earns normal profit.



6-8 MARKS QUESTIONS

1. Explain any four features of perfect competition.
Ans. (i) Very Large Number of Buyers and Sellers: In perfect competition number of firms is very large. No seller is able to influence the market price either by withdrawing from the market or by supplying the entire stock. Like sellers, buyers also cannot influence the price.
(ii) Homogeneous Product: Under perfect competition all sellers sell identical or homogeneous products. They are those products which are identical in quality, shape, design, packing, etc.
(iii) Freedom of Entry and Exit of Firms: A perfect competitive firm has absolute freedom of entry and exit. A firm can enter or leave the industry at any time. There are no legal restrictions on the entry and exit of the firms. A perfect competitive firm earns normal profit in the long run because of freedom of entry and exit.
(iv) Perfect Knowledge about the Market: Buyers and sellers are fully aware of the prices prevailing in the market. Buyers know it fully well at what price sellers are selling a given product. As a consequence only one price prevails in the market.

2. Explain any four features of monopoly market.
Ans. (i) One Seller and Large no. of Buyers: Under monopoly there is only one seller of the commodity. The seller may be in the form of sole proprietorship, partnership or joint stock company but the buyers of the product are large in number.
(ii) No Close Substitutes of the Commodity: A monopoly firm produces a commodity which has no close substitute of it. Pure monopoly will then only exist if there is no close substitute of the commodity.
(iii) Closed or Restricted Entry of the New Firms: Under monopoly there are some restrictions on the entry of new firms in the market for instance there are patent rights as exclusive control the techniques of production or raw material.
(iv) Full Control Over Price: Since there is only one seller in the market, he has full control over price. He is a price maker. He can make variations in the price according to the market conditions.
(v) Price Discrimination: The act of selling the same product at different prices to different buyers is known as price discrimination. Monopolist can easily do price discrimination because he is the only seller of the product in the market. If a monopolist adopts the policy of price discrimination the situation is called discriminating monopoly.

3. Explain four important features of a monopolistic market.
Ans. (i) **Large no. of Buyers and Sellers**: There is large no. of buyers and sellers in the market. However the number of buyers and sellers is less than that prevails under perfect competition.
(ii) **Product Differentiation**: Under monopolistic competition the products of different firms are not homogeneous but are different from each other in terms of brand, name, colour, shape, quality, size, etc. Buyers develop some kind of preferences for the product of a particular seller. Because of product differentiation each firm decides its price policy

(iii) **Selling Cost**: Because of product differentiation each firm in a monopolistic competition has to spend a lot on advertisements of its product in order to sell more. market.
Expenses on advertisement and publicity are called selling costs which play very important role in a monopolistic

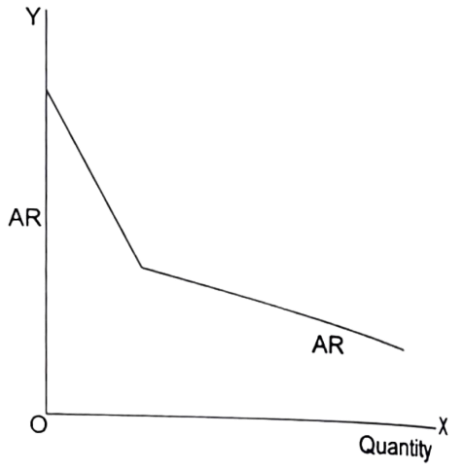
(iv) **Non-Price Competition**: Besides price competition there also exists non-price competition in a monopolistic market, Several firms compete with each other without changing the price of their product. e.g., some firms introduce git schemes with particular purchases, etc.

4. Explain four important features of an oligopoly market.

Ans. (i) **A Few Sellers**: Most important feature of oligopoly is that there are only a few sellers in the market and every seller produces a substantial part of total production of the industry and therefore, every seller is in a position to influence price of the product. The number of sellers can be between 2 to 10.

(ii) **Interdependence of Firms**: Oligopolist firms are interdependent on each other in their decision making. No firms can take independent decision regarding price and output. Each firm will have to take into account the possible reactions of the rival firms.

(iii) **Indeterminant Demand Curve**: An oligopolistic firm can never be certain about the shape of its demand curve. It cannot predict its sales correctly. It is because of the fact that any change in the price and output leads to sharp reactions by its rival firms. As a matter of fact demand curve cannot be precisely determined here. Demand curve in oligopoly is known as Kinked demand curve.



(iv) **Price Rigidity**: Prices are generally stable under oligopoly. Once a price is fixed, it remains same for a long time. It happens because of two reasons:

(i) Each seller might have learnt through experiences the disadvantages of price wars.

(ii) The sellers may go for non-price competition.

5. Explain the **Implication** of following features of various markets.

Ans. (i) **Large number of buyers and sellers in perfect competition market.**

Implications: As there are large number of sellers, individual seller cannot influence market supply or price. Similarly, one buyer cannot affect market demand or price.

(ii) **Homogeneous goods in perfect competition market.**

Implications: (a) Perfect competition market has homogeneous goods which are same in shape, size, colour, price, etc. So it is easy for new firms to enter into and exit from the market.

(b) There is no selling cost as there is no need for advertising the goods. So one firm cannot effect price market that decides the price.

(iii) **Free entry into and exit from the market in perfect competition market.**

Implications: if in short-run there is abnormal profit, firms will enter the market and if there are abnormal losses firms will exit the market.

Hence, in the long-run firms will earn normal profits.

(iv) **Differentiated goods monopolistic competition market**.

Implications: Differentiated goods are different in shape, size, colour, packaging, etc.

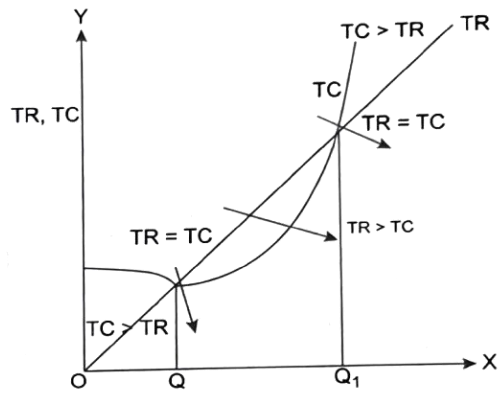
(a) New firms can easily enter the market by adding new feature in their product.

(b) Non-price competition occurs. Firms compete on the basis of different features in goods and not on the

basis of prices.
(c) Selling costs are high as differentiated goods of competitors are close substitutes so firms have to advertise and change the product qualities.
(v) **Downward sloping AR curve in monopoly and monopolistic competition market.**
Implications: Due to downward slope of AR curve and demand curve in both the markets, the firms have to simultaneously decide price and quantity to be sold. If firm decides to sell at a high price then AR curve indicates that less quantity will be sold. If they decide on larger quantity to be sold then AR curve will show the low price to be charged by firms. More goods can be sold at lower price and less goods can be sold at higher price.

6. Explain TR and TC approach to explain producer's equilibrium.
Ans. Total Revenue and Total Cost Approach: As we all know the TR refers to the total sale proceeds that a firm gets from the sale of its output whereas TC refers to total expenditure incurred by firm to produce a given amount of output. Total profit can be calculated by deducting TC from TR thus Profit =TR-TC.

A firm will be in equilibrium when the difference between the two is maximum. This condition of equilibrium of the firm can be explained with the help of graph.



In the adjoining graph TC & TR represent total revenue and total cost curves respectively. The difference between them is measured by vertical distance. At points on the two curves up to OQ level of output, firm is incurring losses, hence TC curve lies above TR curve. Points A and B are break-even points which show the condition of no profit no loss but beyond OQ level of output, firm starts earning profits but the total profit is maximum only at OQ, level of output as the vertical distance between TR & TC curves at this point is maximum. It is also clear from total profit curve which reaches its maximum point at OQ1, level of output.

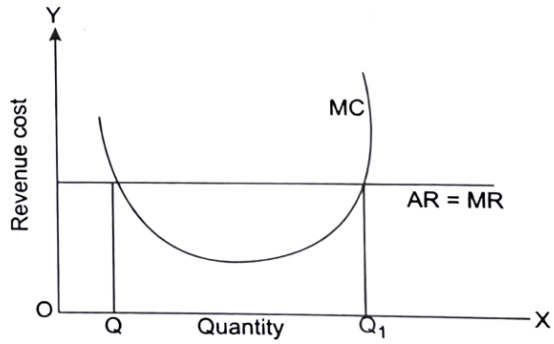
Defects of the TR-TC Approach: Though the TR-TC approach is the simplest approach, yet it suffers from some defects. Its main drawbacks are as under:
1. It is very difficult to find minimum and maximum vertical distance between total revenue and total cost approach.
2. This approach totally ignores the per unit cost of output.
Due to these shortcomings, modern economists have used MR and MC approach in place of TR & TC approach.

7. How does a producer attain equilibrium under perfect competition through MR and MC approach.
Ans. Marginal Revenue and Marginal Cost Approach: MR & MC approach is another method to find out the position of firms equilibrium. As we know MR is a net addition to the TR when 1 more unit of output is sold and MC refers to the addition made to the TC by producing external units of product.

A firm will be in equilibrium when it is earning maximum profit and profits are maximized when MR = MC. Equality of MR and MC is the essential condition but not a sufficient condition of equilibrium that is why the second order condition is that MC should cut MR curve from below. Hence a firm to be in equilibrium the following two conditions must be satisfied:

1. MC = MR
2. MC must cut MR from below.

1. **First Order Condition**: MC = MR. A firm gets maximum profit at a level where MR = MC. If MC <MR, it means firm can earn profit by producing more. On the other hand if MR < MC the additional unit produced by firm reduces profit, therefore it is good for a firm to operate at a level where MR = MC.



2. **A Rule to Ensure that Profits are Maximised Rather Than Minimised (MC must cut MR from below):** MR = MC is the basic rule for profit maximization of a firm but it is not a sufficient condition for it because in certain conditions this equality will minimize the profit instead of maximizing it. It can be explained with the help of following graph:

In the adjoining graph MC curve cuts MR curve at two places A & B but both the points can not be considered as the point of equilibrium because point A gives minimum profit instead of maximum. Point B gives more profit than A and beyond point B, MC> MR it means firm stats incurring losses. Thus, the point B where MC cuts MR from below is the point where firm gets maximum profit.

MR and MC approach is considered to be a superior approach over TR and TC approach because under this approach we can find three things simultaneously:

(i) Equilibrium output,
(ii) Equilibrium price and
(iii) Profit.

8. Illustrate the short-run equilibrium of a perfectly competitive firm incurring losses.

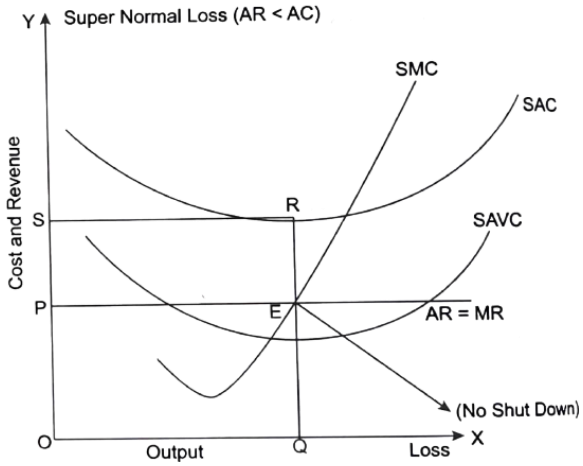
Ans. The general conditions for the equilibrium of the firms in the short-run will be:

(i) SMC = MR
(ii) SMC cuts MR from below.

But these conditions will not guarantee that the profit is earned by the firm. In order to know whether the firm is making profit or loss we need a short-run average cost curve. In this regard there are three conditions for firm's equilibrium.

In short-run:

AR > AC	Super Normal Profit
AR <AC	Loss
AR=AC	Normal Profit.



It is also possible in short-run for a competitive firm to incur losses. In this condition AC > AR & the equation for super normal loss will be:

AR=MR=P=SMC <SAC

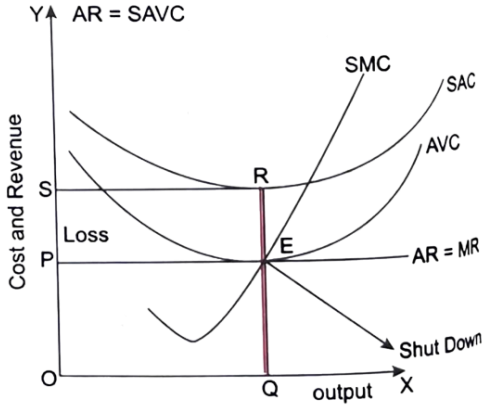
In the above graph TR = OPEQ
TC= OSRQ
TC > TR Thus Loss = PERS

There are two types of conditions in case of super normal losses:

(i) No shut-down: In this case of super normal loss firm will continue its production even if it is incurring losses because it can meet average variable cost easily with the revenue it is earning. In this case,

SAVC <AR <SAC (Shown in the above graph)

(ii) Shut-down Condition: When a firm incurs losses in a short-run it has to decide whether to continue production or not. For this, it has to seek the position of AVC curve in the short-run. If it equals to AR then the firm must stop the production because it shows that the firm is not able to meet its expenditures on its variable factors. Thus, in case of shut-down AR less than or equal to SAVC.



QUESTIONS BASED ON DIFFERENTIATION

1. Differentiate between perfect competition and pure competition.

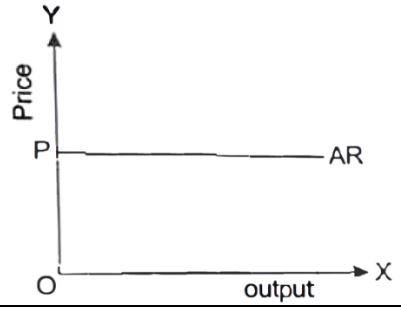
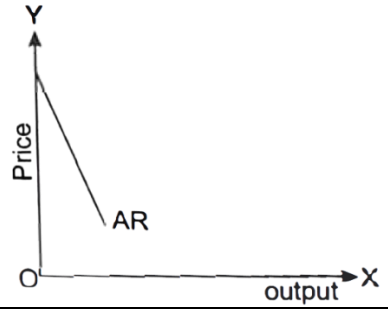
Ans.

S.No.	Basis	Perfect Competition	Pure Competition
1	**Meaning**	It means a form of market where there are large number of buyers and sellers of a commodity which is homogeneous in nature which is sold with no control over price by individual firm.	Pure competition exists when there are a great number of sellers whose products are all identical and no one seller controls a large enough part of the total output to exert an appreciable influence on price.
2	**Scope**	Perfect competition is a wider term and includes pure competition.	Pure competition is a part of perfect competition and is narrow in nature.

2. Differentiate between perfect competition and monopoly

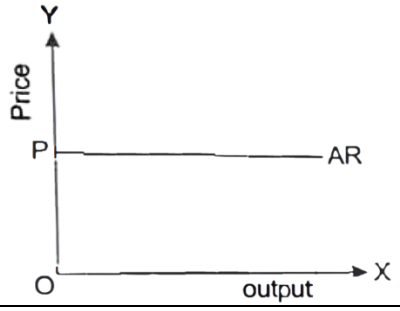
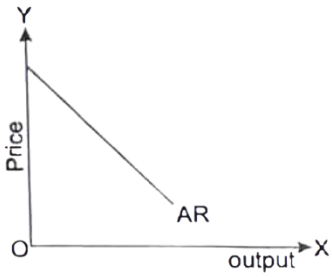
Ans.

S.No.	Basis	Perfect Competition	Monopoly
1	**Meaning**	It means a form of market where there are very large number of buyers and sellers of a commodity which is homogeneous in nature which is sold with no control over price by an individual firm.	Monopoly is a state of market in which there is only one producer or seller of a particular commodity. There is no close substitute of that commodity and no new firm can enter into the market. Monopolist has full control over the price and can also practice price discrimination.

2	**Entry and exit of firms**	Freedom of entry and exit of firms.	Closed/restricted entry and exit.
3	**Nature of the demand curve**	Demand curve of perfect competition is a horizontal straight line parallel to X-axis. 	Demand curve in monopoly market is downward sloping but less elastic. 
4	**Profit in the long- run**	A perfect competitive firm earns normal profit only in the long-run because of freedom of entry and exit.	A monopolist firm earns super normal profit also in the long-run because of restricted entry.

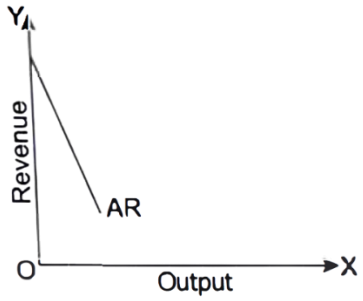
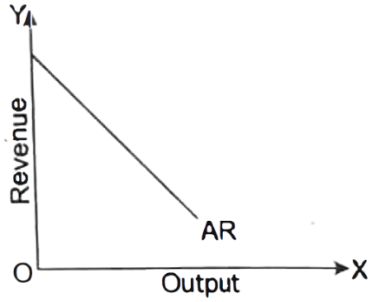
3. Differentiate between perfect competition and monopolistic competition.

Ans.

S.No.	Basis	Perfect Competition	Monopolistic Competition
1	**Meaning**	It means a form of market where there are very large number of buyers and sellers of a commodity which is homogeneous in nature which is sold with no control over price by an individual firm.	Monopolist Competition is a form of market which has large number of sellers and allows the firm the freedon of entry and exut unlike perfect partial control over the price of the product.
2	**Price Policy**	Uniform price policy	Individual price policy
3	**Knowledge about the market**	Perfect knowledge about the market.	Imperfect knowledge about the market.
4	**Nature of the demand curve**	AR curve is a horizontal straight line. 	AR curve is downward sloping. 

4. Differentiate between monopoly and monopolistic competition.

Ans.

S.No.	Basis	Monopoly	Monopolistic Competition
1.	**Meaning**	Monopoly is a state of market in which there is only one producer or seller of a particular commodity. There is no close substitute of that commodity and no new firm can enter into the market. Monopolist has full control over the price and can also practice price discrimination.	Monopolistic competition is a from of market which like perfect competition has large number of seller and allows the firm the freedom of entry and exit and unlike perfect competition, it has product differentiation and partial control over the price of the product.
2.	**Nature of the product**	Product with no close substitute of it.	Differentiated product
3.	**Entry and exit of firms**	Closed entry and exit.	Freedom of entry and exit of firms.
4.	**Selling cost**	Selling cost does not play any role in this market as there is only on product sold in the market and there is no close substitute of it.	Selling cost plays very important role in this market because products sold in this market are substitutes of each other.
5.	**Nature of the demand curve**	Downward sloping but less elastic 	Downward sloping and highly elastic. 
6.	**Control over price**	Full control over price	Partial control over price.

5. Differentiate between monopoly and oligopoly.

Ans.

S.No.	Basis	Monopoly	Oligopoly
1.	**Meaning**	Monopoly is a state of market in which there is only one producer or seller of a particular commodity. There is no close substitute of that commodity and no new firm can enter into the market. Monopolist has full control over the price and can also practice price discrimination.	Oligopoly is a type of imperfect competition. It is a market situation in which there are only a few firms in the industry producing either homogeneous products or closely differentiated products. If there are only two firms it is known as duopoly.
2.	**Number of sellers**	There is a single seller or producer of the commodity.	There are a few sellers or producers of the commodity.

3.	**Product differentiation**	Product differentiation is not possible.	Product differentiation is possible
4.	**Sales promotion activities**	These activities do not play important role.	These activities have an important role to play in this market.
5.	**Profit**	Producers and sellers generally get abnormal profit.	Producers and sellers comparatively get less profit.

CASE BASED QUESTIONS

CASE 1

Oligopoly is a market form. More than a typical market form, uniqueness of oligopoly is that compared to other market models like perfect competition, oligopoly looks a practical market situation.
Oligopoly, by definition is a market where a few sellers producing the product. There can be around 6-9 firms under oligopoly. Hence it is called competition among the few. Under oligopoly, there is intense interdependence among the firms. One firm's action may influence revenue and profit conditions of others. Advertising is an important part of oligopolistic market situation.

Answer the following questions:

1. Give the example of two Oligopoly industries of India.
2. What is the nature of the product in an oligopoly market?
3. What is the nature of the demand curve of an Oligopoly market?
4. Write about the entry and exit of the firms in an Oligopoly market.

Ans. 1. In India, markets for automobiles, cement, steel, aluminium, etc., are the examples of oligopolistic market. In all these markets, there are few firms for each particular product.
2. Oligopoly is said to exist when there are few sellers of homogeneous or differentiated products.
3. There are high barriers to entry and exit of firms in an Oligopoly market. The most important barriers are government licenses, economies of scale, patents, access to expensive and complex technology.

CASE 2

When people hear the term monopolistic competition, they often assume that the term means the same thing as a monopoly, but the two are very different in reality. Monopolistic competition occurs when a market contains a large variety of firms offering very similar products to consumers. The barriers to entry in the market are low, and the decisions made by individual firms rarely affect the other firms. The key to firms positioning themselves in front of consumers is through brand differentiation and advertisement. So unlike a monopoly, where there is only one firm in a market who has total control over product and prices, the firms in a monopolistic competition don't have much effect on each other.

Answer the following questions:

1. Define a Monopolistic Competition.
2. Differentiate between Monopoly and Monopolistic competition on the basis of Price control in the market.
3. What is selling cost? What role selling cost plays in the Monopolistic market and why?

Ans. 1. Monopolistic competition is a form of market which like perfect competition has large number of sellers and allows the firm the freedom of entry and exit and unlike perfect competition it has product differentiation and partial control over the price of the product.

2. Monopoly market has full control over the price as there is only one seller in the market. Monopolistic competition has partial control over the price as there are many sellers selling differentiated products in the market.
3. Because of product differentiation each firm in a monopolistic competition has to spend a lot on advertisements of its product in order to sell more. Expenses on advertisement and publicity are called selling costs., selling costs are essential to push up the sales. They are incurred to persuade a buyer to purchase one product in preference to another.

CASE 3

Monopoly markets are markets where a certain product or service is offered by only one company. A monopolistic market structure has the features of a pure monopoly, where a single company fully controls the market and determines the supply and price of a product or service. Hence, a monopolistic market is a non-competitive market. In a market monopoly, the entry of new firms is not possible in the market in a free manner due to several reasons, including government licences and regulations, huge capital requirements, complicated technology and economies of scale. These barriers to economic growth hinder firms from entering the market.

Answer the following questions:

1. What is the implication of restricted entry and exit of firms in a monopoly market?
2. What is price discrimination?
3. Give the reason for less elastic demand curve in a monopoly market.

Ans. 1. Under monopoly there are some restrictions on the entry of new firms in the market for instance there are patent rights as exclusive control the techniques of production or raw material.
Implications: A monopolist can earn supernormal profit in the long run also.
2. The act of selling the same product at different prices to different buyers is known as price discrimination. Monopolist can easily do price discrimination because he is the only seller of the product in the market. If a monopolist adopts the policy of price discrimination the situation is called discriminating monopoly.
3. Demand curve under monopoly is less elastic as compared to the demand curve under monopolistic competition due to absence of close substitutes in monopoly.

CASE 4

Equilibrium refers to a state of rest when no change is required. A firm (producer) is said to be in equilibrium when it has no inclination to expand or to contract its output. This state either reflects maximum profits or minimum losses. A firm attains the stage of equilibrium when it maximises its profits, i.e., when he maximises the difference between TR and TC. After reaching such a position, there will be no incentive for the producer to increase or decrease the output and the producer will be said to be at equilibrium.

Answer the following questions:

1. What are the two approaches to measure the producer's equilibrium in a perfect competitive market?
2. When does producer get normal profit?
3. What do you mean by a firm?

Ans. 1. TR,TC approach and MR, MC approach.
2. TR=TC and AR = AC.
3. Firm is a single unit of production which employs factors of production to produce goods and services.

QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. Identify the market forms of the following cases:
(i) Goods sold are homogeneous,
(ii) Motor car market in India,
(iii) Market for toilet soaps in India.
Ans. (i) It is a perfectly competitive market because the goods sold in this market are homogeneous in nature.
(ii) The motor car market in India is an example of oligopoly market because of the existence of a few sellers.
(iii) The market for toilet soaps (say, Lux, Nirma, Liril, Dove, Rexona, etc.) in India is an example of monopolistic competition because of the existence of many sellers trying to capture the market share through attractive advertisements.

2. Identify the market forms of the following goods and services:
(i) Railway services in India,
(ii) Paddy sold by the farmers,
(iii) The edible oil sold by its producers,
(iv) Gold ornaments sold by the sellers under different 'brand names',
(v) Personal computers produced and sold by different companies.
Ans. (i) It indicates a monopoly market because railway services are sold only by the Government of India in our country.
(ii) It shows a perfectly competitive market because there are innumerable number of sellers and buyers in the paddy market.
(iii) The edible oil market in India indicates monopolistic competition since a large number of producers or sellers sell their products under different brand names (such as 'Engine', 'Dhara', 'Sundrop', 'Flora', etc.) and try to increase their market shares through non-price competition.
(iv) The gold ornament market in India can also be regarded as monopolistic competition because a large number of sellers sell their products under different brand names such as Senco Gold, "P.C. Chandra Jewellers', 'Nemichand Bamalka', "Tanisk', 'M.P. Jewellers', etc.
(v) The personal computer (PC) market in India can also be considered as monopolistic competition because different companies sell their products under different 'brand names such as HCL. TVS Electronics, LG Electronics, Microtech, Samsung India, HP, Jenith India, etc.

3. Identify the market forms of the following:
(i) Textile industry in India,
(ii) Perfectly elastic demand,
(iii) Telecom industry in India.
Ans. (i) **Textile industry in India**: Monopolistic competition (since there are a large number of industrial units producing textile products in their own brand names, e.g., Vimal, Reymonds, Mayur, DCM, etc.)
(ii) **Perfectly elastic demand**: Perfectly competitive market (since, each firm has to sell at a given price and the demand becomes perfectly elastic at that price).
(iii) **Telecom industry in India**: Oligopoly market (since, only a few public and private sector firms operate in this market with substantial market power, e.g., BSNL, Tata communications, Vodafone, Airtel, etc.)

4. How is a single buyer a price taker in perfect competition?
Ans. A single buyer's share in total market demand is so significant that the buyer cannot influence the market price on his own by changing his demand.

5. Normal profit means zero economic profit. Why?
Ans. Suppose the existing firms are earning above normal profits. Attracted by the positive profits, the new firms enter the industry. The market supply increases and the price comes down. New firms continue to enter and the price continues to fall till economic profits are reduced to zero.
In case of losses, firms start leaving the industry, supply falls and prices starts going up and all this continues till losses are wiped out. Remaining firms in the industry then once again earn just normal profits/zero profit.

6. What is the relation between marginal revenue curve and demand curve under monopolistic competition?
Ans. Both AR and MR curves have negative slope.

7. Under which competitive market can firm earn abnormal profits in the long run?
Ans. Monopoly market.

8. How is the success of price discrimination policy depends on elasticity of demand?
Ans. Monopolist will sell the goods at higher price to the consumer for whom elasticity of demand of a goods is inelastic, whereas he will sell the goods at lower price to the consumer for whom elasticity of demand of a goods is elastic.

9. Why does the marginal revenue curve has negative slope under monopoly?
Ans. Under monopoly AR and MR curves have negative slope because more can be sold by lowering the price.

10. Why is the price line for a competitive firm horizontal?
Ans. Because per unit price of a goods is constant and firms can sell infinite quantity at given price.

11. What is perceived demand curve of a firm?
Ans. When any firm under monopolistic competition, estimates its planned sales on the basis of the assumption that other competitors would not react to changes in the price by that particular firm, that planned sales curve is called the perceived demand curve of that firm.

12. What is a cartel in an oligopoly market? Give an example.
Ans. A cartel is an organization formed through the collusive agreement between the rival firms in an oligopoly market. The member firms of this cartel may agree to sell their product at a common price to avoid a price-war. The Organization of Petroleum Exporting Countries (OPEC) is an example of cartel formed by a few oil producing countries having oligopoly power in the international crude oil market.

13. How does a firm under monopolistic competition exercise partial control over price?
Ans. It happens because by incurring heavy selling cost, the firm is able to create a differentiated image of its product in the minds of consumers. Products are differentiated on the basis of brand, size, colour, shape, etc. Buyers are attracted to buy a particular product even at higher prices.

14. What is patent right?
Ans. Patent right is an exclusive right, granted to a firm to produce a particular product or use a particular technology, as a reward for its risk and investment in research.

15. Why is the demand curve, under monopoly less elastic as compared to the demand curve under monopolistic competition?

Ans. The demand curve, under monopoly is less elastic as compared to the demand curve under monopolistic competition due to the absence of close substitutes of the commodity sold.

WORKSHEET FOR SELF EVALUATION

1. Which features of monopolistic competition are monopolistic in nature? **(2 Marks)**

2. Which features of monopolistic competition are competitive in nature? **(2 Marks)**

3. Explain the nature of demand curve under: (i) Perfect competition (ii) Monopolistic competition (iii) Monopoly (iv) Oligopoly. **(2 Marks)**

4. Why are selling costs not incurred in a perfect competitive market? **(2 Marks)**

5. Differentiate between monopoly and Monopsony. Give an example for each. **(2 Marks)**

6. What is meant by the term price taker in the context of a firm? **(2 Marks)**

7. Justify the following as price takers/ price makers: **(3 Marks)**
(i) An oligopoly market,
(ii) A perfectly competitive market.

8. Why does a uniform price prevails under perfect competition? **(2 Marks)**

9. Explain the implication of the feature product differentiation under monopolistic market. **(2 Marks)**

10. Is it correct to say that profit of a producer under perfect competition is maximum at a level at which P = MC but MC is decreasing? **(2 Marks)**

11. Explain the meaning and conditions of producer's equilibrium. **(3 Marks)**

12. What do you mean by shut-down point? **(3 Marks)**

13. Explain producer's equilibrium with the help of TR and TC curves. **(4 Marks)**

14. Differentiate between shut-down and break-even points with the help of diagrams. **(4 Marks)**

15. Explain the determination of price by an industry in a perfect competitive market. **(3 Marks)**